

Executive Scenario Report

Predicted Load	Uncertainty	Financial Exposure	Grid Status
120.3 MWh	+/- 7.0 MWh	\$16,123	Elevated

Strategic Overview

This is a short-term operational energy forecast predicting a value of 120.32 units for the upcoming period, accompanied by a notable uncertainty of 6.99. The ensemble model, heavily weighted towards CatBoost, indicates a slight stabilization or decrease compared to recent historical averages, influenced by negative contributions from 12-hour and 24-hour lags. The significant spread among individual model predictions suggests underlying model disagreement and potential volatility.

Critical Risk Factors

- High uncertainty: The 6.99 uncertainty value, approximately 5.8% of the prediction, is substantial for an operational forecast, increasing the risk of actual demand deviating significantly and potentially leading to operational inefficiencies or supply-demand imbalances.
- Model divergence: The wide spread between individual model predictions (e.g., CatBoost at 115.51 vs. LightGBM at 131.22) indicates a lack of consensus among the underlying models, which could lead to instability in future forecasts or a less robust ensemble.
- Negative lag influence: The negative contributions from lag_12h and lag_24h suggest a potential downward trend or lower demand compared to recent history, which, if not accurately captured, could lead to over-forecasting if the trend accelerates or under-forecasting if conditions rebound sharply.

Actionable Recommendations

- Implement dynamic hedging strategies: Given the significant uncertainty, operational teams should prepare for a range of outcomes by having flexible generation or demand response mechanisms ready to adjust to actual load fluctuations.
- Investigate model discrepancies: Conduct a deeper analysis into why LightGBM predicts significantly higher than CatBoost and XGBoost, focusing on feature importance and model biases to improve ensemble robustness and reduce prediction variance.
- Monitor real-time trends closely: Pay close attention to real-time demand and weather patterns, especially around the 12-hour and 24-hour marks, to validate the observed negative lag contributions and adjust operational plans if actual trends diverge.

INPUT FEATURE CONTEXT

Hour:	12
Day Of Week:	2
Month:	1
Is Weekend:	0
Lag 24H:	120
Lag 12H:	118
Roll Mean 24H:	125
Roll Std 24H:	4